

Wing Spar Optimization of Radio Controlled Size Aircraft

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Abstract—This honors thesis shall provide a comprehensive optimization of wing spars for use in radio controlled aircraft. The model aircraft chosen for the experimentation is approximately 14 pounds with an approximate wingspan and length of 7 feet. The optimization will compare varying cross sectional spars at differing lengths utilizing several different materials defined by their moduli. The optimization utilizes a Latin hypercube with surrogate modeling. The optimization process has been altered to include a large array of design points for improved results. The results shall provide a basis for wing spar design and provide insight to specific material and geometrical design choices.

I. INTRODUCTION

THE ability to cut weight without the expense of losing structural rigidity is of utmost importance in the design and fabrication of all aircraft. The recent advances in composites, unique alloys, and other complex materials has opened up a world of opportunity for aircraft structural engineers and radio controlled (RC) aircraft designers alike. Available material choices are abundant and each have their own set of benefits and disadvantages. To add greater confusion, material strength is very dependent upon the material's geometries and dimensions. Due to the large amount of alternatives it is helpful to perform a computer based optimization to aide in design and material selection

for the structure of the aircraft.

One of the most structurally intensive parts of the aircraft is that of the wing, its purpose, to provide lift for the aircraft [1]. The wing structure is composed of spars, ribs, stringers, and skin. Spars are situated lengthwise in the wing carrying the majority of bending forces. Ribs lie transversely and provide airfoil shape and minor compression support. Stringers also lie lengthwise and provide support for the skin. The skin covers the internal components of the wing, provides a smooth surface for passing airflow, and handles shear loading. Of all the components in the wing, the spars provide the most significant support and are responsible for transferring the aerodynamic loading, created by the wing, to the aircraft wing box and fuselage [2]. See fig. 1 for an example of wing construction.

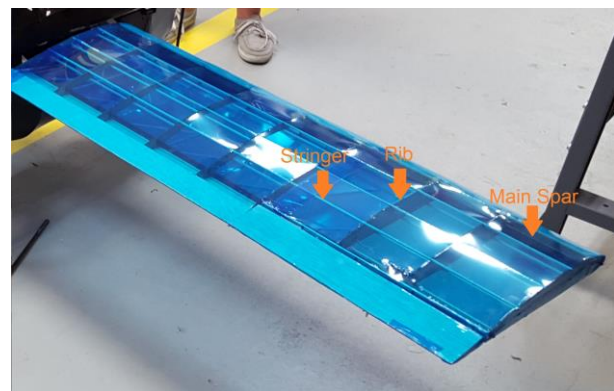


Fig. 1. RC Aircraft Wing Construction

The most common geometrical shapes for aircraft wing spars are I beam, rectangular beam,

and U beam (see fig 2). RC aircraft in particular most commonly use the rectangular beam for its ease to source, manufacture, and incorporate in an RC aircraft's wings. The rectangular shape can vary in aspect ratio (ratio of height to width) altering its moment of inertia and its accompanying resistance to bending forces.

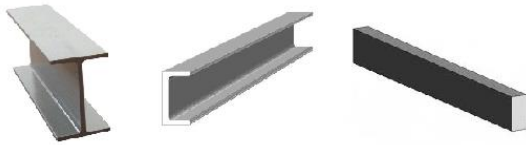


Fig. 2. I beam, U beam, and Rectangular beam respectively.

The most common spar materials found in RC aircraft vary from plastic and wood to carbon fiber and aluminum. The materials each have their own set of material properties, the most relevant being the modulus of elasticity (or Young's Modulus) which measures the resistance of a material to deform. The deformation of the spar is important both in regards to wing tip deflection and yield/failure of the spar. The other driving factor in wingtip deflection is the length of spar itself. The increased length of the wing and its accompanying spar create a longer moment arm contributing to higher stresses and a larger wingtip deflection.

II. PROBLEM STATEMENT

RC aircraft design is often a subjective process especially when determining the best material, size, and length for the main wing spar. Analysis may prove a certain material and size completely capable of performing the job but this is often not the optimal design to maximize strength, minimize weight, and prevent deflection. It can be difficult to choose the best combination of designs due to the expanse of design variables. The optimization performed here aims to determine the best combination of design points for a mid-size 14 pound RC aircraft.

For the sake of this optimization, several assumptions will be made. For one, the aircraft size is assumed to be 14 pounds with an approximate wingspan and length of 7 feet. The size matches several mid-size RC aircraft and is well below the maximum 50 pound limit requirement for RC

aircraft [3]. The optimization thusly assumes this single aircraft design for simplicity in achieving results. It must be noted that a drastically different weight and size could alter the results and may not apply to very large nor very small RC aircraft. The next assumption is in regards to the cross sectional area of the spar. To maintain ease of comparison, width of the cross section will remain constant whereas the height will be altered to increase the cross sectional area for differing designs in the optimization. Furthermore, the elastic modulus for composite and fiber materials will be assumed to be the elastic modulus recorded in the "1" direction, E_1 . Finally, it is assumed that the original spar design is based off a previous analyzation (which in fact it is). The initial/original design utilized in the optimization will be a unidirectional carbon fiber rod with a cross sectional area of 0.200 in^2 and a length of 96 in. This was the selected spar for the University of Florida's Design Build Fly team and was chosen based off preliminary analysis and no optimization.

III. ALGORITHM OF OPTIMIZATION

The optimization employed for the experiment first begins with defining a function. The function is known as the cost function, J . Each term in the cost function has a design parameter associated with it. These terms can have added multipliers to better categorize their importance in the design. The parameters utilized here are based off the major importance in the design of the wing spars. They include weight, deflection, and factor of safety (FOS). The parameters are dependent upon the design variables. The design variables include cross sectional area, modulus of elasticity, and the length of the spar. A range of values is required for each design variable and is chosen by selecting the most realistic maximum and minimum values for the design. Length, for example, is limited at the high end by the ability to transport the aircraft (for example must fit inside a transportation vehicle) and at the lower end due to the importance of increasing a wing's aspect ratio (to improve lift to drag ratio and maintain a sufficiently long wing). The ranges utilized are 0.15 to 0.40 in^2 for cross sectional area, 435 to 23200 ksi for modulus of elasticity, and 80 to 120 in for length.

The resulting cost function includes each of the parameters mentioned and adds multipliers to relate the terms closer together. Even so, emphasis is highest with weight (even though it is scored with the same multiplier, the weight values had a greater difference between the design points and the original), followed by deflection, and then FOS. The FOS term is unique due to the desire of meeting an FOS of 1.5 (the standard for most aircraft components) [4]. Thus the resulting FOS is divided by the desired FOS and subtracted by 1. The absolute value of that result is utilized in case the FOS goes above or below the target. FOS's that result in anything less than 1 will be automatically disqualified (the cost will be manually adjusted to a very large value). FOS is also multiplied by 50 due to its very small initial contribution to the cost function (See eqn. 1).

$$J = 3 \frac{(Weight)_{DP}}{(Weight)_{orig}} + 3 \frac{(Deflection)_{DP}}{(Deflection)_{orig}} + 50 \left| \left(\frac{FOS_{DP}}{1.5} \right) - 1 \right| \quad (1)$$

The optimization begins with inputting the minimum and maximum ranges for each variable. A Matlab code created by Dr. Richard Lind is utilized for the optimization albeit in modified form [5]. The initial code is set up to use a Latin hypercube to develop initial design points. The code is altered to utilize 50 points in the design space versus 5 and then handpick 5 points evenly spread out amongst the distribution. This allows for a greater resolution of varying design points aimed at avoiding local minimums in the data (and helps to find the absolute minima). While using a greater amount of design points has the advantage of finding more accurate results, it has the disadvantage of increased cost (in the form of time). The amount of time to run the code more than doubles. For the sake of this experiment, the 50 design points provide a good balance between computational time and accuracy.

After the initial 5 design points are found, they are used to alter a Solidworks model [6]. Each new model is placed into a fundamental element analysis (FEA) simulation study. A wingtip loading of 5 times the force of gravity is used to simulate an aggressive turn maneuver likely to be seen in flight (equivalent to 70 lbs_f). The simulation provides the parameters to input in the cost function. The cost function is evaluated at each of the original 5 points

and the results are then input into a second code which performs surrogate modeling. This process is repeated two more times until there are 200 total points in the design field and 20 evaluated points.

The FEA employs a tetrahedral mesh TET10 element with 4 Jacobian points. A mesh density of "coarse" was selected for improved speed during analyses.

The results are tabulated and compared. The optimal design is the one that minimizes the cost function.

IV. ANALYSIS OF RESULTS

An original optimization was performed outside of this thesis experiment utilizing only 20 total points and the results were not very refined. For this thesis, the optimization was retuned and improved vastly. This result can be proven by judging the differences in the costs in Table I.

Table 1		
Design Point	Cost (Original)	Cost (New)
1	7.835	30.29
2	16.58	14.38
3	17.21	93.76
4	250.4	93.87
5	48.79	57.49
6	25.34	35.61
7	25.24	84.85
8	26.27	11.22
9	26.31	12.96
10	70.10	81.44
11	718.9	122.0
12	697.8	25.67
13	659.8	33.77
14	613.2	131.2
15	575.6	58.10
16	20.72	44.77
17	20.17	92.25
18	21.39	45.21
19	20.70	42.59
20	20.20	12.87

Note the wildly varying (and high) costs of the first experiment compared with this current experiment. The 20 point optimization proved to be less than ideal compared to the 200 point optimization. One similarity between the two optimizations (not shown) was that the original

design still was the most optimal design (showing that the initial analysis to derive the original design was rather accurate). The reason for improvement for data also was due to changes made to the cost function. The ranges for the variables were adjusted to improve the optimization. The time utilized to carry out the optimization could be seen as wasted time but the results show some rather interesting trends (in addition to proving the correct material/size spar was chosen). See Table 2 for the design points that were evaluated. See Table 3 for the evaluated costs.

Table 2				
Design Point	Weight (Ounces)	Deflection (in)	FOS	Cost (J)
Original	17.6	6.7	1.51	6.44
1	35.0	7.5	2.13	30.29
2	23.6	13.6	1.37	14.39
3	30.0	5.3	0.41	93.77
4	7.0	100.1	0.07	93.88
5	57.1	4.2	0.12	57.49
6	24.2	1.8	3.62	35.61
7	30.2	1.1	5.08	84.85
8	22.6	2.4	2.89	11.23
9	15.1	8.3	1.37	12.96
10	25.6	1.3	4.54	81.45
11	24.8	1.1	6.22	122.04
12	59.2	1.4	1.05	25.68
13	34.4	1.8	3.51	33.77
14	24.8	0.67	6.5	131.22
15	28.7	1.5	4.27	58.10
16	18.3	12.1	0.41	44.77
17	5.4	99.3	0.095	92.25
18	93.2	10.4	0.75	45.21
19	20.1	50.6	1.23	42.59
20	44.2	5.5	1.40	12.87

Table 3			
Design Point	Area (in ²)	Modulus (ksi)	Length (in)
Original	0.197	19580	96.0
1	0.385	4152.4	105
2	0.206	23206	120
3	0.318	6941.5	93.1
4	0.267	1364.8	118
5	0.344	8799.4	98.8
6	0.288	23206	88.0
7	0.352	23206	90.0
8	0.263	23206	90.0
9	0.175	23206	90.0

10	0.312	23206	86.0
11	0.162	23206	98.0
12	0.387	23206	98.0
13	0.225	23206	86.0
14	0.162	23206	86.0
15	0.187	23206	96.0
16	0.213	12959	100
17	0.213	1573.6	100
18	0.213	14097	98.0
19	0.213	2712.2	98.0
20	0.213	15236	84.0

The major trend showed that materials with a modulus less than 100 (such as pine wood, spruce wood, hemp fiber, and fiberglass) were far and away too weak for the proposed size aircraft. On the other end, the materials with higher modulus (and an accompanying increased weight) were simply too heavy and/or “overdesigned” for the role (this included a carbon fiber impregnated plastic, heavier and stronger than standard carbon fiber laminates). The only acceptable designs with the larger modulus were those with small cross sections and shorter wing spans (to alleviate the weight issue), and unfortunately had too much deflection.

The optimal design ended up being the initial unidirectional carbon fiber rod measuring 96 inches in length. The resulting FEA is shown in figures 3, 4, and 5.

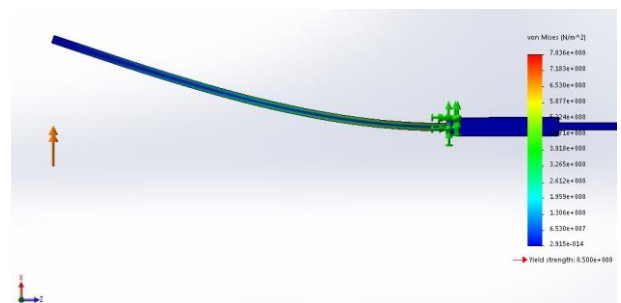


Fig. 3. Wing tip loading showing stress

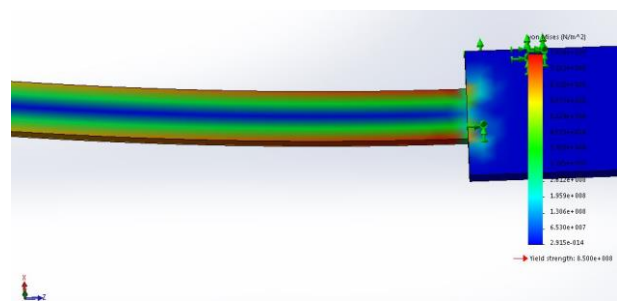


Fig. 4. Zoom in of the area with the highest stress concentration

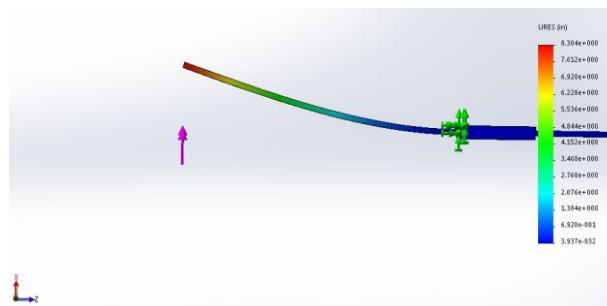


Fig. 5. Image of the deflection in the spar with distance

Taking a closer look at figure 4 it can be seen that the majority of the stress is localized near the wing root. The root must deal with counteracting the moment produced at the wingtip and the upward force seen through the body. The deflection seen in figure 5 is maximum at the tip as is completely expected. Although the wing will see a distributed load the single point load serves as an accurate approximation. Wing tip tests are very common in industry and utilized by the large aircraft manufacturers, Airbus and Boeing [7].

Inferring the results and viewing the simulation data provides some serious insight into available improvements both in the design and optimization process.

For the design, there are several opportunities to improve the spar's abilities. One advantage could be to utilize a tapered spar (increasing from the tip to the wing box) to allow sufficient resistance to the higher loads at the root and decrease mass at the tips. For the sake of this experiment all spars were uniform (since they are easier to source and manufacture), but for the sake of the best design this is one valid design change. In addition, instead of using a single material or one type of laminate, a sandwich construction could be utilized. Sandwich construction consists of utilizing a lightweight/soft material like balsa wood sandwiched between two thin layers of very strong material like carbon fiber. The advantage is decreased weight without the severe penalty of a drastically reduced modulus of elasticity. Finally, the spar could be made hollow (like a box). While pricier to source, the design would cut down weight and maintain sufficient strength.

For the optimization there are also areas of weakness. For one, the length range should be

shortened. The reason for the larger range (80 to 120) was to satisfy mainly satisfy aerodynamic "wants." A larger wing and shorter chord is most favored by the aerodynamicist as it decreases drag and increases lift. Unfortunately, a structure has to make it all the way to the wingtips and there are limits to the possible overall length. The longer the wing, the stronger the structure needs to be (especially if the aircraft is carrying a payload), and the heavier the resulting aircraft. If the range were brought down lower it would allow greater optimization to instead occur with the cross section and the material choices. In addition, the optimization could still use a greater amount of design points. The more numerous the design points the greater the chance there is to find a minimum in the data. Finally, the optimization could also be improved by altering the FOS term in the cost function. For many design points a large FOS was harming their overall cost when in fact the large FOS was not actually a negative aspect of the design (so long as the weight was not too great). The FOS is mainly utilized to ensure the wing spar would not become "overdesigned," but there were times when this hampered a design's success by too great of a margin.

V. CONCLUSION

RC aircraft are a unique take on aircraft design. While smaller than real aircraft (and not as expensive/sensitive to accidents) they still require time to design, analyze, and fabricate. While RC planes can be designed, built, and flown by hand, they may not achieve much more than a nosedive or structural failure. During the design of larger RC aircraft this realization becomes even more important. Deeper analysis and optimization can improve designs, provide successful builds, and offer enjoyable flight time. This thesis attacked the problem of selecting an optimal wing spar for a small to mid-size payload carrying RC aircraft. The experiment began with a previously analyzed spar and sought to improve upon the design (or at least provide insight into possible design changes). What resulted was a large amount of useful information for both the physical design and the optimization process.

Surprisingly, the best wing spar ended up being

the original spar design. This begged the question: was all that time for optimization worth it? The time taken to optimize was quite less than the time taken to conceptualize and analyze the original design and provided quite a large amount of data to consider. The optimization process provided helpful alternatives and aided in ensuring the best design was chose. It more than proved its usefulness in the aircraft design process.

As has been very evident throughout engineering curriculum, sometimes the results are not as desired and often not as expected. It is important to localize the reasons for these results, infer the data, and apply, alter, or expand the next round of testing to resolve any issues discovered. An engineer does not merely perform a test and call it a day, rather, they apply and implement the principles and lessons learned to improve designs. Application is king.

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